Model Parameter Tables

Table 1: Group 1: well constrained parameters, fixed in all runs.

Symbol	Description	Value	Units	Reference	Notes
ρ_f	Seawater (fluid) density	1.0275	$\mathrm{g\,cm^{-3}}$	Millero and Poisson [1981]	*Standard seawater. Millero & Poisson provide the international equation of state; value changes* $< 0.1\%$ *over North Atlantic conditions.*
ν	Kinematic viscosity	0.01	$\mathrm{cm^2\,s^{-1}}$	Sharqawy et al. [2010]	*Seawater at 20 °C. Varies ∼30% between 5 and 25 °C; currently held fixed (temperature scaling for future work).*
g	Gravitational acceleration	980	$\mathrm{cm\,s^{-2}}$	—	*Standard value; invariant for ocean column depths of interest.*
k_B	Boltzmann constant	1.3×10^{-16}	$\mathrm{erg\,K^{-1}}$	—	*NIST fundamental constant; enters only the Brownian kernel.*
a_w	Settling speed coefficient (kriest_8)	66	$\mathrm{m\,day^{-1}\,cm^{-b_w}}$	Kriest [2002]	*Power-law fit to in-situ Atlantic settling data (Eq. 8 of Kriest 2002). Coefficient and exponent chosen together; treat as a pair.*
b_w	Settling speed exponent (kriest_8)	0.62	—	Kriest [2002]	*As above. Exponent is robust across datasets; coefficient varies by ∼factor 2 between cruises.*
$\Delta\rho_{fp}$	Fecal pellet excess density	0.15	$\mathrm{g\,cm^{-3}}$	Turner [2002]; Ploug et al. [2008]	*Dense compact pellets (∼150* $\mathrm{kg\,m^{-3}}$ *above seawater). Gives ∼69* $\mathrm{m\,day^{-1}}$ *at bin 8 via Stokes law, consistent with Turner (2002) review and Ploug et al. (2008) measurements.*

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Table 1 continued...

Symbol	**Description**	**Value**	**Units**	**Reference**	**Notes**
d_0	Monomer (unit particle) diameter	20×10^{-4}	cm	Burd and Jackson [2009]	*20 μm monomer, typical phytoplankton cell. Sets the size grid; changes shift all bin boundaries.*
A_D	$D_{\max}$ amplitude in $D_{\max} = A_D\,\varepsilon^{-1/4}$	9.39×10^{-6}	$\mathrm{m^{3/4}\,W^{-1/4}}$	Burd and Jackson [2009]	*From Kolmogorov turbulence theory. $D_{\max}$ sets the largest stable aggregate size per depth layer. Amplitude is better constrained than the exponent.*

Table 2: Group 2: parameters with reasonable literature estimates but site-dependent values. Held at literature defaults in the first optimization pass; fit in a second pass.

Symbol	**Description**	**Value**	**Units**	**Reference**	**Basis / confidence**
D_f	Particle fractal dimension	2.33	—	Logan and Wilkinson [1990]; Li and Logan [1995]	*Ocean aggregates: range 1.7–2.5 across studies. Value 2.33 is a mid-range choice consistent with marine snow measurements. Controls mass–size relationship and kernel collision cross-sections.*
r/r_g	Ratio of collision radius to radius of gyration	1.6	—	Li and Logan [1997]	*Sets effective collision cross-section for shear and differential settling kernels. Li & Logan (1997) derive this from fractal geometry; value varies ∼1.3–2.0.*
Z_c	Filter feeder (macrozooplankton) concentration	0.307	$\mathrm{ind\,m^{-3}}$	Stemmann et al. [2004a]; Stemmann et al. [2004b]	*Atlantic maximum from Stemmann et al. (2004) Table 2. Depth profile from Fig. 1 wired into `DepthProfile.typical()`. Needs EXPORTS net tow data (Amy / Debbie) for refinement.*
c	Filter feeder clearance rate	0.025	$\mathrm{m^3\,ind^{-1}\,day^{-1}}$	Stemmann et al. [2004a]	*From Stemmann et al. (2004). Order-of-magnitude known; exact value uncertain.*
Z_f	Flux feeder concentration	0.063	$\mathrm{ind\,m^{-3}}$	Stemmann et al. [2004a]; Stemmann et al. [2004b]	*Atlantic maximum from Stemmann et al. (2004) Table 2. Same depth profile logic as Z_c.*
s	Flux feeder capture cross-section	1.3×10^{-5}	$\mathrm{m^2\,ind^{-1}}$	Stemmann et al. [2004a]	*Stemmann et al. (2004). Interpreted as the projected area presented by a sinking particle to a flux feeder.*

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Table 2 continued...

Symbol	Description	Value	Units	Reference	Basis / confidence
p	Egestion fraction (fecal / ingested)	0.3–0.5	—	Stemmann et al. [2004a]	*Range reflects food quality and organism type. Default 0.5 (current). Controls fecal pellet production; linked to carbon budget closure.*
r_0	Microbial remineralization base rate	0.03	day^{-1}	Iversen and Ploug [2013]	*Iversen & Ploug (2013) measured respiration on sinking diatom aggregates in a Norwegian fjord. Value represents warm surface conditions; deep column rates* $\sim$*10*$\times$ *lower after* Q_{10} *scaling.*
Q_{10}	Temperature sensitivity of microbial rate	2.0	—	Iversen and Ploug [2013]	*Standard marine bacteria value. Doubles the rate per 10 °C increase. Widely used; uncertainty* ± 0.5.
s_m	Micro-zooplankton mining capture cross-section	1.3×10^{-5}	$\mathrm{m}^2\,\mathrm{ind}^{-1}$	Stemmann et al. [2004a]	*From Stemmann et al. (2004) Part I Eq. 25; same value as flux feeder cross-section.*
f_{next}	Fraction of disaggregated biovolume sent to next smaller bin	2/3	—	Burd and Jackson [2009]	*Sectional model redistribution convention. Remainder spread uniformly across smaller bins. No direct measurement; value is a modeling assumption.*

Table 3: Group 3: parameters with no reliable direct field estimate. These are fit first against EXPORTS UVP particle size distributions (North Atlantic station, surface to 500 m). Grid search over each parameter individually; conjugate gradient or simulated annealing for joint optimization if needed.

Symbol	Description	Current value	Units	Reference	Why unknown / fit range
α	Collision efficiency (stickiness) of marine snow aggregates	1.0	—	Jackson [1990]; Burd and Jackson [2009]	*No direct measurement for natural marine particles. Jackson (1990) set* $\alpha = 1$ *(upper bound). TEP-coated aggregates may have* $\alpha \sim 0.01$–1.0*; value depends on surface chemistry and particle type.* ***Fit first****: grid search* $\alpha \in [0.1, 1.0]$*, step 0.1.*
α_{cross}	Stickiness between fecal pellets and marine snow	0.5	—	—	*No field or lab measurement exists for fecal pellet – marine snow collisions. Fecal pellets are compact (*$D_f \approx 2.8$*) and low in TEP, so assumed less sticky than aggregates. Current 0.5 is a conservative starting value.* ***Fit after*** α*; range [0.1, 1.0].*
δm	Biovolume removed per encounter (micro-zooplankton mining)	10^{-5}	cm^3	Stemmann et al. [2004a]	*Proxy for gut volume of small copepods (Oncaea spp.). Stemmann et al. (2004) Part I Eq. 25 gives the functional form but does not constrain* δm *directly. Range* 10^{-6} *to* 10^{-4} *cm*^{3}*. Needs EXPORTS gut-content data (Amy / Debbie).* ***Fit with*** α_{cross}*.*

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Table 3 continued...

Symbol	Description	Current value	Units	Reference	Why unknown / fit range
Z_m	Micro-zooplankton miner concentration	250	ind m^{-3}	Stemmann et al. [2004a]	*Approximate value from Stemmann et al. (2004) Fig. 1. Depth profile $Z_m(z)$ is currently set uniform; needs EXPORTS net tow data to set the actual profile. Uncertainty large ($\times 2$–5).*

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